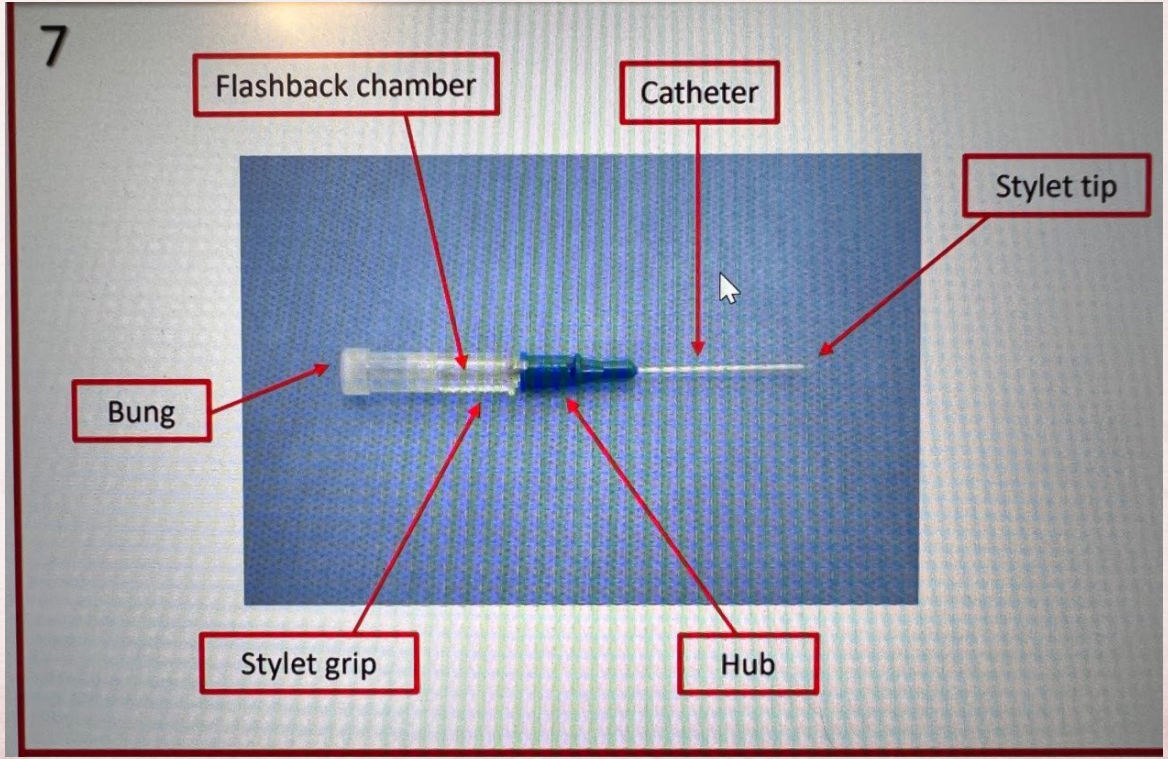
Intravenous Catheters

What is an IV Catheter?

Catheter "anatomy"

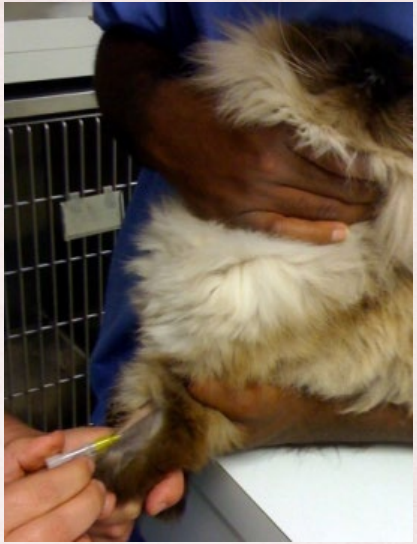


Why do we place IV catheters?

- Purpose: Provide vascular access
  - Usually need temporary venous access for medications, fluids and electrolyte replacement therapy
  - Can be used for transfusion of blood products
  - Emergency IV access

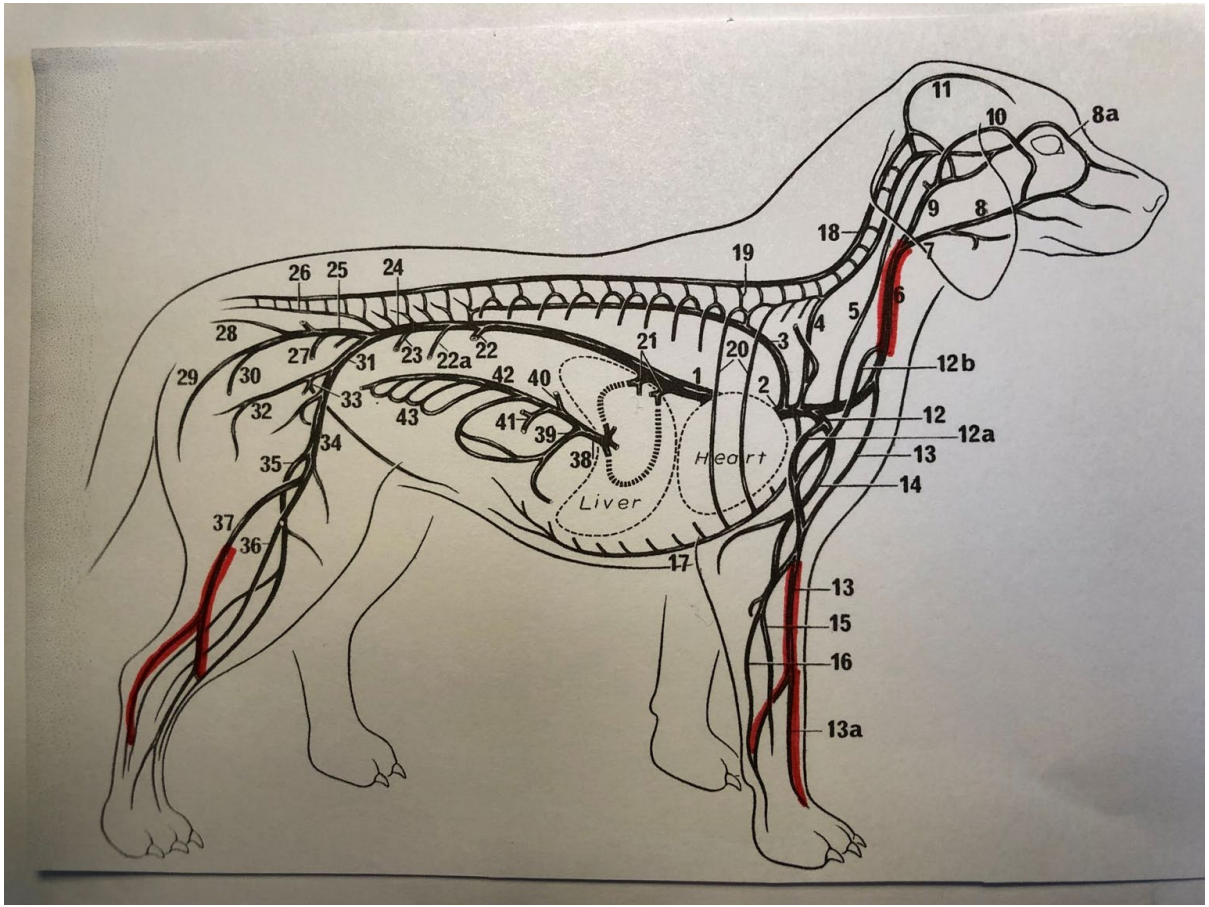
Where do you place an IVC?

- Mostly use the common peripheral venipuncture veins
- Do not use jugular unless placing a jugular catheter
- Try to use an area that is easiest to tape in
  - Always start as distally as possible



Veins for Canine IV Catheterization

- ❑ Cephalic
- ❑ External jugular
- ❑ Lateral saphenous



Considerations when choosing an IVC

1. Need to have sufficient length and secure it properly

2. Choose a large diameter catheter in case rapid fluid administration is necessary

3. Place in a location that will not interfere with any procedure that needs to be performed

4. Be sure to use an administration set with an injection port so medication can be administered

5. Take care to not occlude the catheter when securing the patient to the surgical table (if applicable)

6. BP cuff placement

Types of Catheters

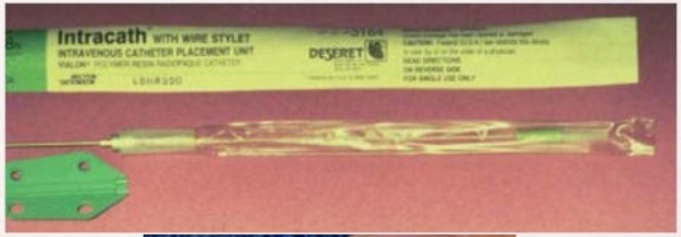
- Common types of IVC:
  1. Through-the-needle
  2. Over-the-needle
  3. Winged needle (butterfly)
  4. Multi-lumen catheter

Over the Needle

- Most common type of catheter used today
- Catheter is fitted outside or over a steel needle
- The needle tip extends past the catheter about a millimeter
- Easy to place
- Less traumatic
- Relatively inexpensive
- 1-2 days

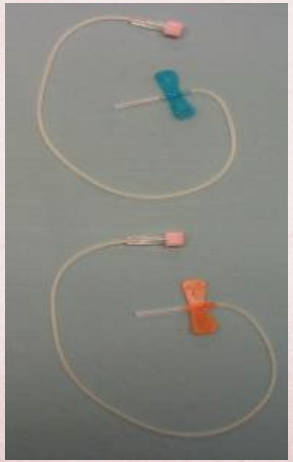
Through the needle

- Used most commonly in critical care patients
- Used for central lines- jugular vein or caudal vena cava
  - Strict aseptic technique
- Passed through or inside the needle
- Are longer than over the needle catheters and have a range in diameters
  - Hypertonic solutions
- Complex
- Expensive
- Time consuming
- 4-5 days with proper care

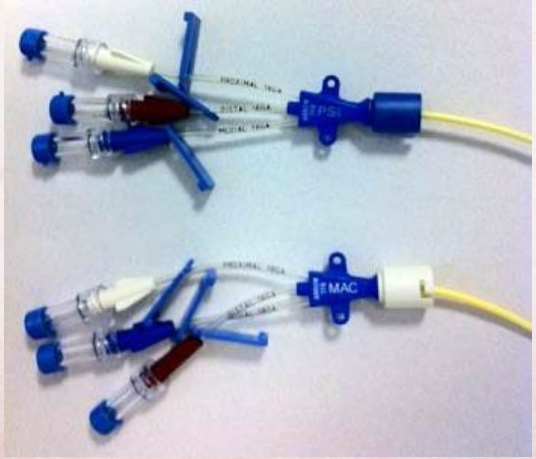


- Winged needle (butterfly)
  - Short term use- needles have plastic wings the facilitate placing or taping in
  - Can lacerate the vein

- Multi-lumen catheter
  - Two to three separate lumens in one catheter- allow for simultaneous infusions in a single catheter site



Wire guided



Catheter Placement

- Common catheter sites include the Cephalic, Saphenous (medial- cat, lateral- dog)
- Wash your hands or wear gloves
- Shave the site- must use 40 blade at minimum
- Aseptically clean the site with antiseptic scrub (3 scrubs)
- Can use a facilitative incision or relief hole to reduce skin tension and friction against the catheter
- Start distal on the limb in case you need to try repeat attempt more proximal

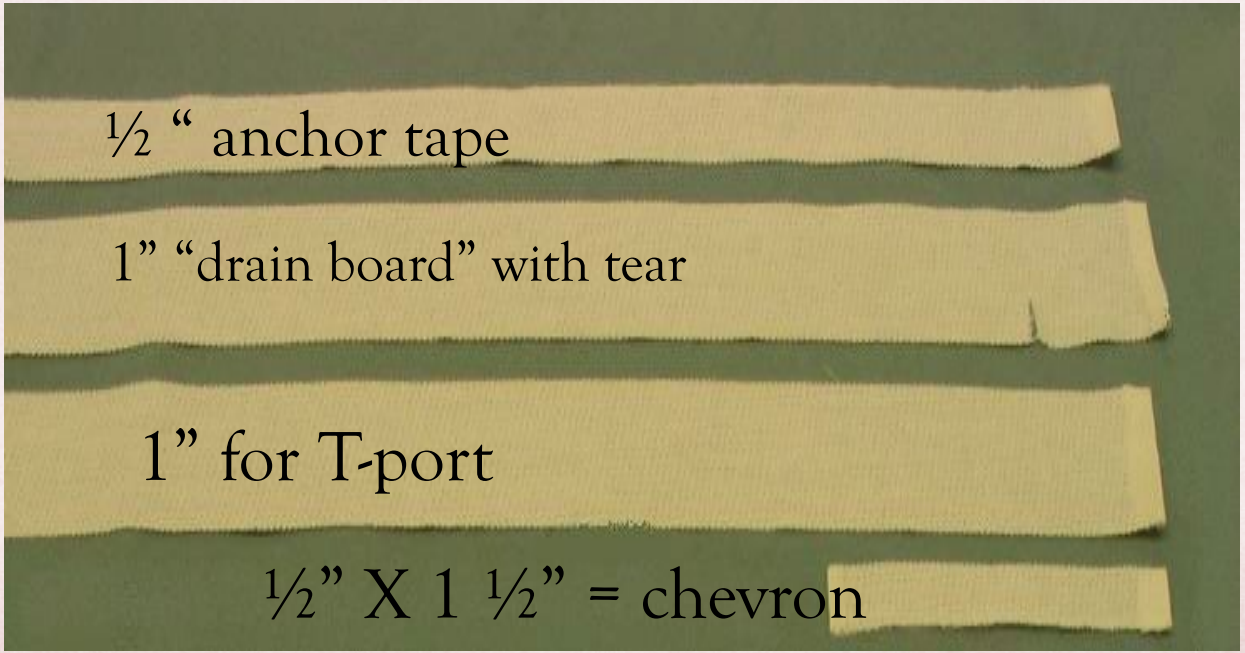
Catheter placement

- Aseptic introduction is critical to avoid contamination and possible infection
- Never reinsert the needle if you fail to thread the catheter. Major problems can arise
- $20\text{-}30^0$ angle
- Brisk motion
  - Too slow push vein out of the way
  - Too fast puncture through vein
- WATCH FOR FLASH

Preparation – Tape Selection

- Porous Tape

Length = 2 – 3 times diameter of limb



Catheter Taping

1. Anchor is very important for security.
   Usually secured with ½” tape sticky side up

2. “Drain board” sticky side down

3. Chevron tape to secure t-port, sticky side up,
   but with twist = ? (difficult to describe)

4. Fluid line securing tape, sticky side down

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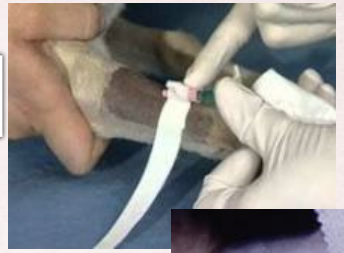

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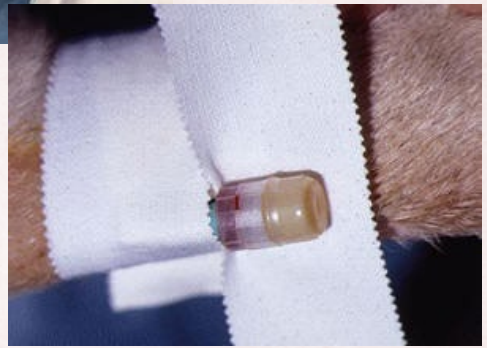

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Catheter Maintenance

- Changing the catheter every 48-72 hours = old thinking..
- Unwrap and check puncture site at least every 24 hours, best to do every 12 hours.
- Heparin lock- consider if not using the catheter to prevent clotting
  - Heparin Saline recipe: 250-500IU heparin to 250 ml normal saline. (1:1 or 2:1 solution)
  - Alternate technique: Can simply flush with normal saline more frequently

Complications - Catheter

- Occlusion
  - Clot formation
  - Kink
- Malpositioning
  - Location of the vessels is unknown
  - Taping of the catheter is not optimal.
  - Movement
- Burring or fraying of the catheter
- Breaking the catheter
- Embolism

Complications - Patient

- Thrombophlebitis
- Phlebitis
- Extravasation
- Thrombus formation
- Infiltration of subcutaneous tissues
- Hemorrhage
- Subcutaneous hematoma formation
- Systemic problems may involve pyrogenic reaction, circulatory overload, air embolism, or allergic reaction

A note about IV catheters for chemotherapy

- IV catheters for cytotoxic solutions
  - Must be a “clean stick”
  - Perform frequent patency checks
    - Watch for signs of extravasation
    - Correction of extravasation

The End
Ian McAllister